Week 04: Moral Dilemmas, Pluralism and Social Choice

Machine Ethics and Artificial Intelligence Safety

Machine Ethics and Artificial Intelligence Safety (11118BLG001). Prof. Dr. Utku Kose. Full Professor, Department of Computer Engineering, Süleyman Demirel University, Isparta, Türkiye. Additional affiliations: University of North Dakota (USA), Universidad Panamericana (Mexico City, Mexico), Vel Tech University (Chennai, India). ORCID 0000-0002-9652-6415. E-mail: utkukose@sdu.edu.tr, utku.kose@und.edu, ukose@up.edu.mx, utkukose@gmail.com. This course is updated in line with current developments in the field. First version: September 2026. Last update: October 2026.

Overview

If machines should reflect human values, whose values count and how are they combined? This week examines dilemma studies such as the Moral Machine experiment and the mathematics of aggregating preferences, including results that show why no aggregation rule is neutral [1, 2].

Learning outcomes

By the end of the week, students are expected to interpret the findings and limitations of large dilemma studies. They should be able to compute plurality, Borda and Condorcet outcomes for a preference profile and to explain Arrow's impossibility theorem and majority cycles. Finally, they should be able to argue when the aggregation of preferences is an appropriate basis for machine behaviour.

Lecture

Dilemmas as research instruments

Moral dilemmas are simplified scenarios in which every option has a moral cost. The trolley cases of Foot and Thomson became the template [3, 4]. Vasant and Kose used dilemma scenarios to compare the possible actions of humans and AI-based systems, showing how the same scenario raises different questions when the agent is a machine [5]. Bonnefon, Shariff and Rahwan found a social dilemma in attitudes towards autonomous vehicles [6]. Respondents approved of vehicles that would sacrifice their passengers to save more pedestrians. Yet they preferred to ride in vehicles that would protect their passengers.

Dilemmas are useful because they isolate one variable at a time. They are also artificial. Real systems rarely face a clean choice between two certain outcomes, and a system designed around dramatic dilemmas may neglect the many small trade-offs that matter more in daily operation.

The Moral Machine experiment

Awad and colleagues deployed an online platform that presented unavoidable accident scenarios and collected about forty million decisions in ten languages from people in 233 countries and territories [1]. Strong global preferences emerged for sparing humans over animals, more lives over fewer and young lives over old. Preferences also varied across cultural clusters, which the authors linked to cultural and economic variables.

The study is a landmark and a cautionary tale. Its participants were self-selected, its scenarios were stylised and it measured stated rather than revealed preferences. More fundamentally, a majority preference is not automatically a justified norm. A preference for sparing people of higher status, for example, raises questions of equal treatment that a vote cannot settle.

Aggregating preferences

Noothigattu and colleagues proposed a voting-based system that learns a model of each participant's preferences from dilemma responses. When a new decision arises, the system aggregates these models with a voting rule [7]. The approach inherits the properties of social choice theory, which studies how individual rankings can be combined.

Social choice theory contains sobering results. Condorcet observed that majority preferences can be cyclic: A majority may prefer option A to B, another majority B to C and a third C to A [8]. Arrow proved that no rule for aggregating rankings of three or more options satisfies a small set of reasonable conditions at once [2]. The conditions are unrestricted domain, the Pareto principle, independence of irrelevant alternatives and non-dictatorship. Different rules therefore produce different winners from the same preferences. Plurality counts only first choices, the Borda count rewards broad acceptability and a Condorcet winner, when one exists, beats every rival in pairwise majority votes. Figure 4.1 shows how often no Condorcet winner exists when rankings are diverse.

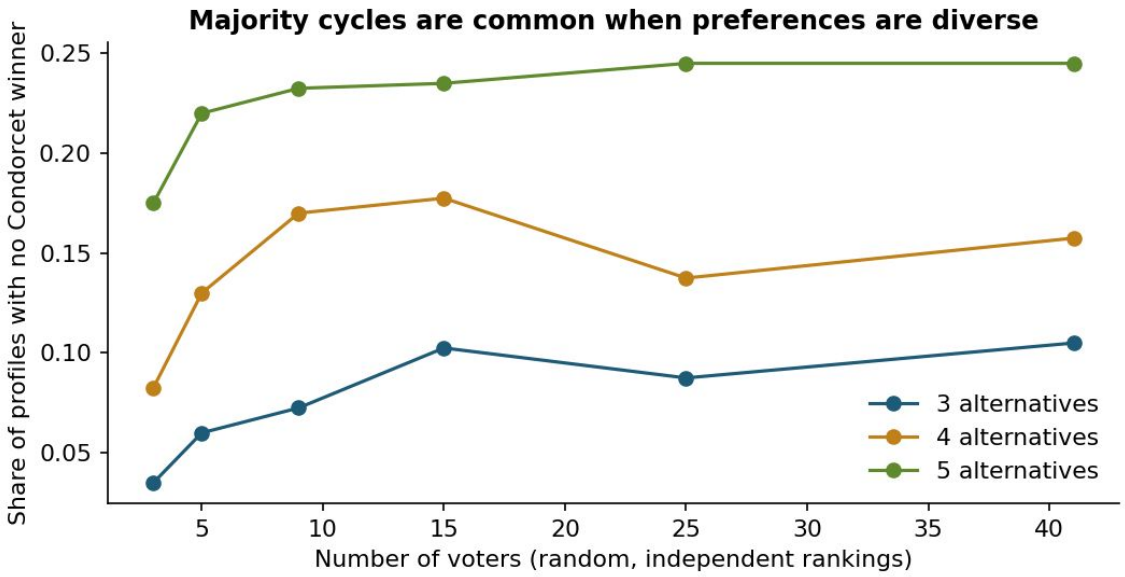


Figure 4.1. Simulated share of preference profiles without a Condorcet winner when every voter ranks the alternatives independently at random. The chance of a majority cycle grows with the number of alternatives [2, 8].

Pluralism beyond aggregation

Aggregation treats morality as a matter of preference. Gabriel argued instead for principles that people could endorse despite their disagreements, selected through a fair process that protects minorities [9]. Moral uncertainty offers a further perspective, in which a system weighs theories rather than voters [10]. In practice, designers can combine the approaches: Hard constraints protect rights that no majority may override, and aggregation is used only among options that respect those constraints.

> ***Pause and reflect.*** *Should the behaviour of a medical triage system deployed in Türkiye follow the preferences of Turkish respondents, of global respondents or of neither? What would justify each answer?*

Hands-on components

Interactive lab: Part A records choices in six accident dilemmas and summarises them as a preference profile. Part B combines a ranking with those of a synthetic class of forty voters and compares plurality, Borda and pairwise majority outcomes [1, 7]. Link: https://utkukose.github.io/machine-ethics-ai-safety-NB-lecture/weeks/week-04/lab.html

Colab notebook: It implements plurality, Borda and Condorcet rules and estimates how often majority cycles occur. It then aggregates synthetic dilemma preferences in the spirit of the voting-based approach of Noothigattu and colleagues [7]. Link: https://colab.research.google.com/github/utkukose/machine-ethics-ai-safety-NB-lecture/blob/main/weeks/week-04/NB04_dilemmas_social_choice.ipynb

Reflection prompts

1. Did your ranking in Part B match the class winner under any rule? What does that suggest about building a system from aggregated preferences?
2. Name one preference revealed in dilemma studies that should not be implemented even if a majority holds it, and explain why.
3. How would you collect moral preferences for a system in your field so that the sample is appropriate and minorities are protected?

Weekly task and submission

What to do. Design a small study that would ask people how an AI system in your field should decide. First, write four dilemma items, each with two options. Second, define who would answer and how they would be sampled. Third, name the rule that would aggregate the answers,

such as simple majority or a supermajority, and explain the choice. For items with more than two options, the Borda count is a further choice. Fourth, list the constraints that no result of the vote may override.

Where to look. Lecture: sections "The Moral Machine experiment", "Aggregating preferences" and "Pluralism beyond aggregation". Interactive lab: Part A for answering dilemmas and Part B for comparing voting rules. Colab notebook: Section 1 for the voting rules and Section 3 for aggregating synthetic dilemma preferences.

Example. For a tool that allocates hospital beds, one item could compare two patients. Should a younger patient with a lower chance of survival be preferred to an older patient with a higher chance? One constraint could state that no aggregation may rank patients by ethnicity or income.

What to submit. A study design of about one page, and the Colab notebook with Exercises 1 and 2 completed.

The weekly task supports self-learning and builds a personal portfolio. In an active semester, students who follow the course with the instructor can send their work for evaluation to utkukose@sdu.edu.tr or utkukose@gmail.com. The e-mail contains the items that the task lists under What to submit. It also includes the learning log exported from the interactive lab (tab: Reflect and submit).

References

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